

M – Tech Production & Marketing

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REPORT OF EXPERIMENT 9

Deep Learning — Neural Networks & Backpropagation

Introduction

This report summarizes the activities completed during **Experiment 9** of the M-Tech Production & Marketing AI/ML Internship Program. The experiment focused on understanding the fundamentals of **Deep Learning**, including Artificial Neural Networks (ANN), forward propagation, backpropagation, activation functions, and building a simple neural network using TensorFlow/Keras. The practical implementation provided hands-on experience in training and evaluating a deep learning model.

Objectives

- To understand the basic concepts of Deep Learning.
- To learn the structure of Artificial Neural Networks (ANN).
- To understand forward propagation and backpropagation.
- To build a simple neural network using TensorFlow/Keras.
- To train and evaluate a neural network model.
- To analyze prediction accuracy.

Task Description

The assigned task was to develop a Deep Learning model using TensorFlow and Keras. A sample dataset was loaded and divided into training and testing sets. A neural network consisting of input, hidden, and output layers was created, compiled, and trained using the Adam optimizer. After training, the model was evaluated, and

predictions were generated to understand the performance of neural networks in classification problems.

Work Done

Today, I developed a simple Artificial Neural Network using TensorFlow and Keras. I loaded the Iris dataset, prepared the training and testing data, designed a neural network with multiple dense layers, and trained the model using the Adam optimizer. After completing the training process, I evaluated the model's accuracy and generated predictions on test samples. Finally, I analyzed the results to understand how deep learning models learn patterns and perform classification tasks.

Challenges Faced

During the experiment, I encountered some issues related to installing the TensorFlow library and configuring the Python environment. These issues were resolved by verifying the required packages, checking the Python version compatibility, and reviewing the program until the implementation was correctly understood.

Key Learnings

This experiment enhanced my understanding of Deep Learning and Artificial Neural Networks. I learned about neurons, hidden layers, activation functions, forward propagation, backpropagation, model training, and performance evaluation using TensorFlow and Keras. These concepts provide a strong foundation for developing intelligent AI applications.

Conclusion

The experiment was completed successfully, and all objectives were achieved. The practical implementation improved my understanding of neural networks and the complete Deep Learning workflow. This experiment strengthened my confidence in building, training, and evaluating Deep Learning models using Python and TensorFlow.

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